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Simulation of Wave Damping Near Coast due to Offshore Wind Farms

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ABSTRACT

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Two hundred wind turbines with an annual productivity of 2.3 TWh, which could produce up to 30% of the energy Estonia needs, are scheduled to be constructed on separate shallows in the NW Estonian coastal waters (the Baltic Sea), 5–20 km off shore. With numerical modeling, we have established a potential impact of the wind farms on wave heights. We concluded that the impact exists in terms of the reduction of significant wave height, but it is very marginal, not more than 1% below 10-m isobaths. This is due to a very small ratio between the turbine diameter and dominant wavelength and the favorable setup of turbines with respect to each other and the coast.

ADDITIONAL INDEX WORDS: *Wind turbines, wave damping, SWAN, Baltic Sea.*

INTRODUCTION

The ever growing need for energy and the decreasing availability of nonrenewable energy sources has put the use of alternative energy sources at the forefront. Apart from nuclear energy, which now generates 6.5% of world's energy, renewable energy—solar, geothermal, water, wind, and biomass—is steadily increasing, now generating up to 13.1% of global energy consumption (IEA, 2007). As renewable energy supplies increase, we begin to look not only toward the increasing need for energy, but toward the reduction of environmental effects as well; climate change due to excess CO_2 is a well-documented fact (IPCC, 2007).

In Estonia (Figure 1) up to 5.1% of energy will be produced from renewable energy sources by the end of 2010 (European Commission, 2007). This number will grow in the future, since nearly 200 wind turbines are scheduled to be installed in the shallows near Hiiumaa. The total power of these arrays would be 600–1000 MW, and the productivity 2.3–2.5 TWh/year (Nelja Energia, 2010). This is about 30% of the total Estonian electricity requirement, maximally. In comparison, in the leading wind energy country, Denmark, only 19.7% of the domestic need is covered by wind energy (Danish Energy Agency, 2007). In addition to the aforementioned 200 turbines, which will most likely be installed in coming years, another 500 MW of arrays could be installed in some shallows in the coastal waters of the northern Estonia as well (Erm *et al.*, 2009).

Every structure that is installed in water will affect the wave and circulation regime and the sediment transport. There are plenty of studies published in scientific literature assessing the importance of scour. Only a handful of studies assess the impact of wind turbines or wave farms upon surface waves near the coast (CEFAS, 2005; DHI Water and Environment, 2007; Millar, Smith, and Reeve, 2007). In scientific papers, however, no studies are published that assess the importance of the whole array of wind turbines upon wind waves. The purpose of this study is to quantify the effect of these structures on wind waves, which propagate toward the coast. The main aim is to find out how much these structures will absorb and scatter wave energy.

Wind farms are scheduled to be installed in the coastal waters of Estonia near Hiiumaa, the NE Baltic Sea. Basically, when planning offshore wind farms, they should be located some distance away from the shore to reduce the effect on land. Construction costs can be reduced if an offshore wind farm is located in shallows. Water is relatively deep off the coast of Hiiumaa, but there are several shallows where water depth is less than 20 m.

This study focuses on two locations—wind farm 1, situated WSW from the Kõpu Peninsula (Figure 1), and wind farm 2, situated NW from Hiiumaa (Figure 1). Coastal areas that could be affected are the head of Kõpu Peninsula (where the so-called surf paradise is located) and the northern coast of Hiiumaa. At wind farm 1, 55 turbines will be installed in parallel rows with a minimum distance of 1000 m from each other. The closest turbines at the wind farm 1 will be 5 km from the coast (the head of Kõpu Peninsula) and the farthest ones 15 km from the coast of the Kõpu Peninsula. The nearest coast from wind farm 2 is the western part of Tahkuna Peninsula, which is 13 km to the SE. The northern coast of the Kõpu Peninsula directly south of the wind farm 2 is 20 km away.

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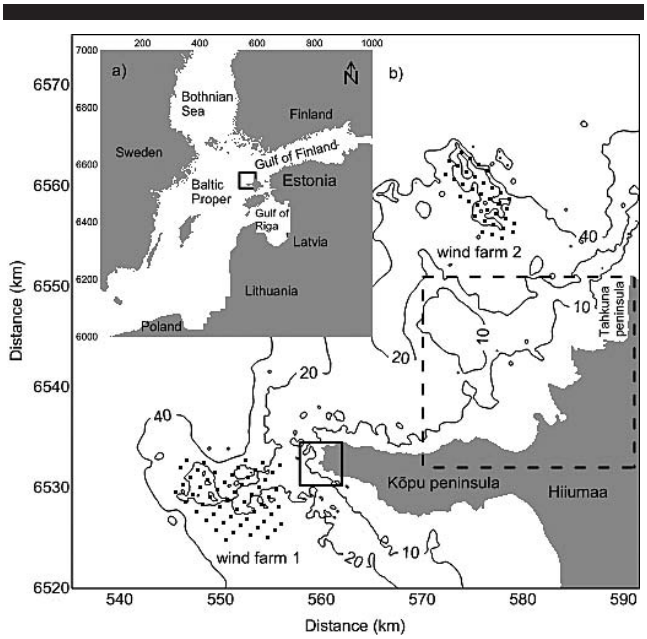

Figure 1. (a) The Baltic Sea, its subbasins and countries residing. The black box is the boundary of Figure 1b. Note that (a) represents the 2000-m grid as well. North up; (b) A detailed location of the wind farms. The black continuous box represents the impact area of wind farm 1 and the dashed black box represents the impact area of wind farm 2. The contour levels are the water depth in meters. Every black-filled circle represents a wind turbine. North up.

In Section 2 the model is described, set up, and validated. Main findings are presented and discussed in Section 3; key conclusions are outlined in Section 4.

METHODS

Model Description

The Simulating Waves Nearshore (SWAN) model is used to assess a potential effect of wind farms upon wave heights. SWAN is a third-generation phase averaged spectral wave model developed at the Delft University of Technology, the Netherlands (Booij, Ris, and Holthuijsen, 1999). In SWAN, waves are described with a two-dimensional wave action density spectrum, whereas the evolution of the action density, N, is governed by the time-dependent wave action balance equation, which in Cartesian coordinates reads

$$\frac{\partial N}{\partial t} + \left(\vec{c}_g + \vec{U}\right)\cdot\nabla N + \frac{\partial c_\sigma N}{\partial \sigma} + \frac{\partial c_\theta N}{\partial \theta} = \frac{S_{\text{tot}}}{\sigma}. \quad (1)$$

The first term represents the local rate of change in action density; the second term denotes the propagation of wave energy in the two-dimensional geographical space with $\vec{c}_g$ the group velocity, and $\vec{U}$ the ambient current. The third term represents the effect of shifting the relative frequency due to variations in depth and mean currents. The fourth term represents the depth-induced and current-induced refraction. The quantities c_σ and c_θ are the propagation velocities in spectral space (σ, θ), with σ and θ representing the relative frequency and the direction of propagation, respectively. The right-hand side contains the source term S_{tot} that represents all physical processes that generate, dissipate, or redistribute wave energy. In shallow water, six processes contribute to S_{tot}

$$S_{\text{tot}} = S_{\text{wind}} + S_{\text{nl3}} + S_{\text{nl4}} + S_{\text{wc}} + S_{\text{bot}} + S_{\text{db}}. \quad (2)$$

These terms denote, respectively, the energy input by wind (S_{wind}), the nonlinear transfer of wave energy through three-wave (S_{nl3}) and four-wave interactions (S_{nl4}), and the dissipation of waves due to white-capping (S_{wc}), bottom friction (S_{bot}), and depth-induced wave breaking (S_{db}).

Model Setup

The model (version 40.72) was run in the third-generation mode with the saturation-based whitecapping model, which estimates the whitecapping of wind sea locally in the wave spectrum and therefore gives realistic estimates of peak periods (Van der Westhuysen, Zijlema, and Battjes, 2007). In every model run, 40 frequencies distributed logarithmically in the range of 0.05–1 Hz were used and 24 spectral directions were included. The diffraction was activated only when the highest spatial resolution was used.

Steady wind speeds of 8 m/s and 15 m/s were selected. The first value is the annual average wind speed offshore (Soomere, 2003), while the second represents strong wind events that occur at least once per month in autumn and spring (Soomere and Keevallik, 2003). All presented wave fields are saturated. In the case of wind farm 1 winds from the SW and west were selected. The head of the Kõpu Peninsula is located NE from wind farm 1. Thus, waves generated by winds blowing from the north, east, and south do not propagate toward the coast. Even the waves coming from the NW do not make landfall at the head of the Kõpu Peninsula.

Wind farm 2 is orientated so that the waves from the north and NW could reach the coasts of Hiiumaa. Thus, the impact of wind farm 2 is assessed for winds blowing from those directions.

The modeling of waves at wind farm 1 has been carried out using five consequently nested models. First the whole Baltic Sea with a 2000-m resolution is modeled. With the boundary conditions from the previous model, waters surrounding Hiiumaa are modeled with a 400-m resolution, and then 100, 50, and 25 m. The same scheme applies also for wind farm 2, but the highest model resolution is 50 m, not 25 m. The details of the grids are listed in Tables 1, 2, and the coarsest model boundaries visualized in Figure 1.

The Baltic Sea bottom topography by Seifert, Kayser, and Tauber (2001) and the marine charts provided by the Estonian Maritime Administration were used to create the model topographies. The model grids are all Cartesian, and the coordinate system used is Universal Transverse Mercator (UTM-34). The scattered data were interpolated to the model grid. The land mass of Hiiumaa was blanked using a digitized coastline that was extracted from Google Earth. Water depths lower than 0.1 m were replaced with a value of 0.1 m. In the case of the 2000-m model, some small bays in the southern Baltic Sea were artificially masked to reduce the model calculation time. It has no effect on the wave heights near

Table 1. *The location of the origin of the computational grid, number of meshes of the computational grid, mesh sizes of the computational grid for wind farm 1 impact calculations.*

	2000 m	400 m	100 m	50 m	25 m
Location of origin of x direction (km)	50	524	535.2	541	543
Location of origin of y direction (km)	6000	6500	6510	6521	6524
No. meshes in x direction	475	140	348	420	760
No. meshes in y direction	500	150	400	380	480

Hiiumaa. Model bathymetry was smoothed at the boundaries of nested model grids to avoid spurious waves.

For every forcing situation, two calculations are made—first without farms and second with farms. For every wind turbine, the closest point is located in the grid and the value at that grid point is replaced with 0, *e.g.,* it is represented as land. In SWAN the land absorbs all incoming wave energy. Here we assess the changes in significant wave height. Let Hs_1 be the calculation made without farms and Hs_2 be the calculations with farms in nodes (i, j). The impact is defined

$$\Delta Hs(i,j) = \frac{Hs_2(i,j) - Hs_1(i,j)}{Hs_1(i,j)} \times 100\%. \quad (3)$$

If $\Delta Hs(i, j) < 0$, the impact is in terms of reduction, and when $\Delta Hs(i, j) > 0$, it shows an increase. The diameter of the windmill structure (monopile) is not more than 5 m in reality. Assuming a linear dependence between the differences of calculated height, in the case of a 25-m grid step, the results obtained with Eq. 3 are divided by a factor of 5, and in the case of a 50-m grid size, results are divided by a factor of 10.

Model Validation

Subsurface pressure sensors were deployed in the locations of wind farms 1 and 2 in the summer of 2007. Although the pressure gauges remained in the water for 23 days, the first 10 days were selected for the comparison. The second half of the measurements displayed a very low wave activity. Subsurface pressure was converted to surface spectra using the method described by Alari, Raudsepp, and Kõuts (2008).

The model was forced with high-resolution limited area model (HIRLAM) wind fields (resolutions $1' \times 1.6'$ in latitudes and longitudes respectively). During the verification period, wind speed did not exceed 12–13 m/s and wind blew mainly from the SW–west.

The temporal variability of the measured significant wave heights and peak periods are similar at both locations (Figures 2 and 3). The mean significant wave height and the mean peak period were greater at wind farm 1 compared with wind farm 2 (Table 3). The waves generated by the SW winds refract around the Kõpu Peninsula and lose energy. It results in a lower significant wave height and peak period at wind farm 2 compared with the location of wind farm 1. The simulated significant wave height matches the measurements equally well at both locations. The root mean square errors (RMSEs) are 0.2 m and 0.23 m, respectively, but the scatter in terms of height is smaller at wind farm 1 (Table 3). There are time slices when significant wave height is better represented at one location than at the other. The timing of period is somewhat better represented at wind farm 2, but the scatter is basically the same for both sites. Usually, the model fails to reproduce peak periods in both locations synchronously.

Table 2. *Same as Table 1 but for wind farm 2 impact calculations. The 2000-m grid has the same setup as described in Table 1.*

	400 m	100 m	50 m
Location of origin of x direction (km)	560	566.4	570
Location of origin of y direction (km)	6532	6532	6532
No. meshes in x direction	90	256	420
No. meshes in y direction	110	380	700

RESULTS AND DISCUSSION

It is helpful to recognize significant wave height and wavelength in the vicinity of wind turbines prior to the construction. Since the field distribution of the parameters are quite homogenous at the wind farm areas, the results are summarized in Tables 4 and 5, instead of in figures.

For wind farm 1, although there is a fourfold increase of wave heights in the case of doubled wind speeds and a linear increase of wave period, the 45° turn of the wind does not change the parameters at any significant level. For the SW wind, significant wave height is greater only by 0.2 m compared with the west wind at wind farm 1. The peak period is only 0.4 seconds greater. The resulting wavelengths display the same variability. Since the fetch length for the west winds is 250 km and for the SW winds 650 km, almost the same

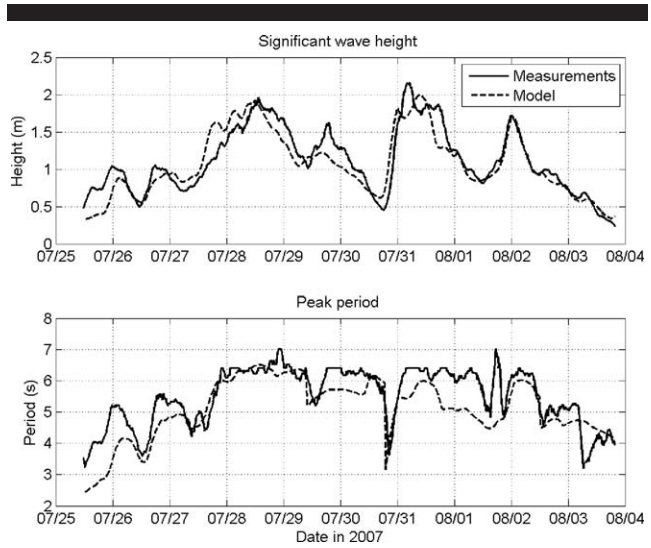

Figure 2. Model-data comparison at the location of wind farm 1.

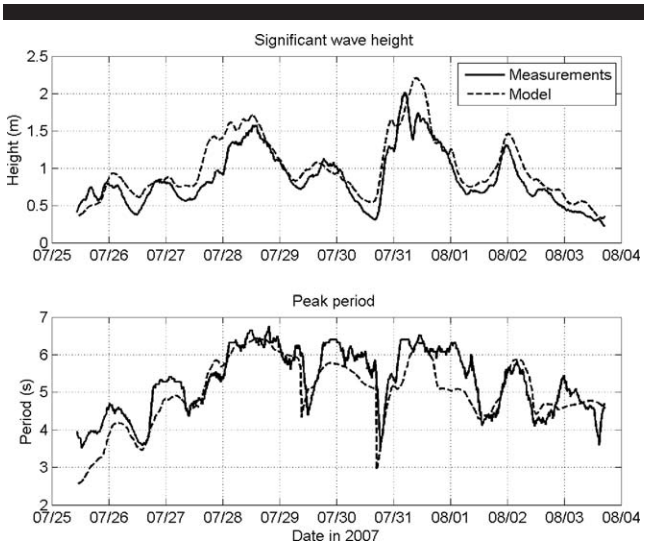

Figure 3. Model-data comparison at the location of wind farm 2.

variability means that the wave field is already full developed at 250 km.

At wind farm 2, the turning of wind from the NW to the north reduces significant wave height up to 0.5 m and peak period up to 1 second. The NW wind generates higher waves since the fetch is longer. However, compared with the wave parameters at wind farm 1, the modeled values of wave parameters are lower at wind farm 2. This is expected since the fetch length is lower in case of wind farm 2.

As an illustration, a field distribution of significant wave heights in the case of 15 m/s, the SW winds are presented in Figure 4. The variability on significant wave heights and wavelengths at the farm locations suggests that dissipation due to bottom friction occur in areas where significant wave height is lower than the mean value, but where wavelengths stay at the mean value. In places where significant wave height is greater than the mean value and wavelengths shorter than the mean value, shoaling exceeds dissipation. The 15 m/s wind induces greater dissipation and shoaling compared with the 8 m/s wind.

In terms of significant wave height, the absolute reduction near the coast due to the wind farm 1 is not more than 2%

Table 3. *The statistics of the validation of the wave model.* R *is the correlation coefficient, scatter index (SI) is defined as the ratio between RMSE and the mean of measurements.*

	Wind farm 1	Wind farm 2
Water depth (m)	14	11
Data points	1350	1350
Hs mean (m)	1.12	0.86
Peak mean (s)	5.56	5.27
RMSE *Hs* (m)	0.2	0.23
RMSE Peak (s)	0.7	0.58
R *Hs*	0.88	0.9
R Peak	0.81	0.86
SI *Hs*	0.18	0.27
SI Peak	0.13	0.11

Table 4. *Modeled wave parameters at the wind farm 1 location.* Hs—*significant wave height; Peak—peak wave period; Wlen—wavelength corresponding to peak wave period.*

	SW 8 m/s	SW 15 m/s	West 8 m/s	West 15 m/s
Hs (m)	0.9–0.95	3.2–3.8	0.88–0.94	3–3.6
Peak (s)	4.7–4.8	8.5–8.7	4.7–4.8	8.1–8.3
Wlen (m)	33–36	75–110	32–35	70–100

(Figure 5). The 1% isoline is closer to the coast in the case of the 8 m/s wind compared with the 15 m/s wind. The difference in terms of significant wave height does not exceed 0.5 cm and 1 cm near the coast in the case of the 8 m/s and 15 m/s wind, respectively. In the case of the 15 m/s wind, the 1% reduction line coincides with the 10-m isobath. Compared with the impact of wind farm 1, the reduction of significant wave height due to wind farm 2 is even more marginal (Figure 6). The reduction is less than 0.25% below the 10-m isobath.

A very low impact is due to three factors. First, waves meeting an obstacle reflect and bend around it. Only large piles change the wave field considerably. Depending on an author (see, for example, Coastal Engineering Manual 2003), the diffraction and reflection are valuated as significant for $D/L > 0.1$ or $D/L > 0.2$, where D is the diameter of the monopile and L is the wavelength. In case of the 8 m/s wind, the ratio is between 0.1 and 0.2, and in case of the 15 m/s wind, it is between 0.05 and 0.08 (when the diameter is 5 m, wavelengths from Tables 4 and 5). Thus in our cases, diffraction and reflection are negligible. Second, the two turbines are not closer than 1 km to each other (at least 10 crests fit between) and therefore very little scatter occurs. And last, the wind turbines are distant from the coast (Figure 2).

A 25-m grid step might be a slight overkill for practical forecasts and hindcasts, and one might think that the results obtained with such a high resolution are not valid anymore, since the grid size is up to four times lower than the peak wavelength. However, when comparing the output, significant wave height, and wavelength with the output of the 100-m and 400-m grid, the results do not deviate (not illustrated here). The dissipation and shoaling is captured as well (of course, the 25-m grid shows more detail—Figure 4).

A question arises: is the linear scaling between 25 m to 5 m reliable? Do we get the same results with different scaling, say from between 100 m and 5 m? We also performed a numerical experiment with the 100-m grid, when the diameter of the turbine was 100 m. After scaling it down to 5 m, *e.g.,* dividing the impact by a factor of 20, we obtained the same result: in the case of the 15 m/s west wind, the reduction was 1% at 10-m isobath. However, the 100-m case displayed some 0.5% to 1%

Table 5. *Modeled wave parameters at the wind farm 2 location.* Hs—*significant wave height; Peak—peak wave period; Wlen—wavelength corresponding to peak wave period.*

	NW 8 m/s	NW 15 m/s	North 8 m/s	North 15 m/s
Hs (m)	0.85–0.9	2.8–3.2	0.82–0.84	2.5–2.7
Peak (s)	4.4–4.6	7.4–7.7	4.1–4.2	6.5–6.7
Wlen (m)	29–31	62–84	26–28	52–66

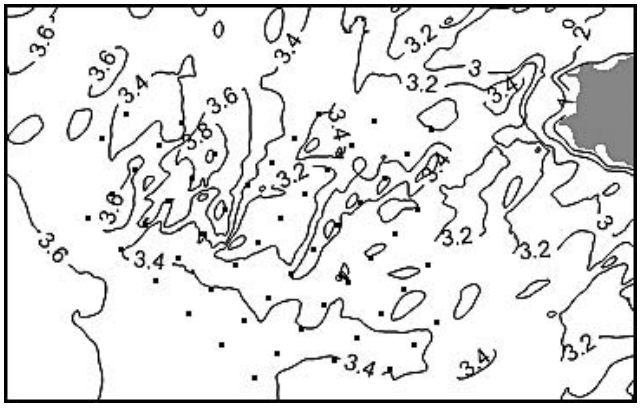

Figure 4. The field distribution of saturated wave field of significant wave height at wind farm 1. The contour unit is in meters. Every black-filled circle represents a wind turbine.

increase of the significant wave height grid point next to shore, which in reality might not exist.

The model results of this study are directly comparable with other findings from other researchers and consultancies. The loss in significant wave height due to Scorby Sands offshore wind farm was less than 2% (CEFAS, 2005). In the latter case a model that was based on an evolution equation solution to the mild slope equation for water waves was used. It differs from the spectral wave model. The model grid size in their experiment was 3 m. Millar, Smith, and Reeve (2005) used the SWAN model in order to assess the impact of a wave farm upon wind waves. In their study the interaction of surface waves and wave farms were modeled using the so-called transmission coefficient. Applying a realistic transmission coefficient, the total reduction of significant wave height near the coast was less than 2 cm. The wave damping due to Rodsand offshore wind farm was less than 1% for higher waves (DHI Water and Environment, 2007). In the study made by DHI (2007), the energy loss was calculated with the integral wave model WAMIT for a single turbine. The impact of the whole array was further established with the spectral wave model MIKE 21 SW.

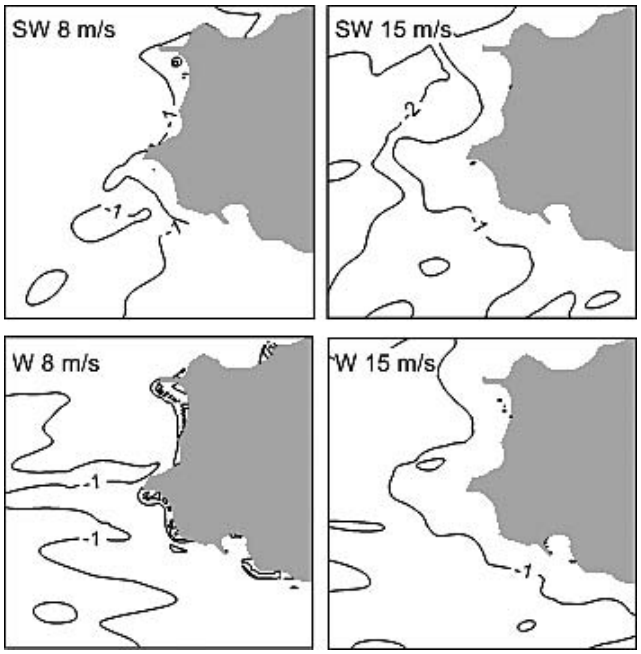

Figure 5. The impact of wind farm 1. The contour unit is [%] and a negative value shows a decrease of significant wave height.

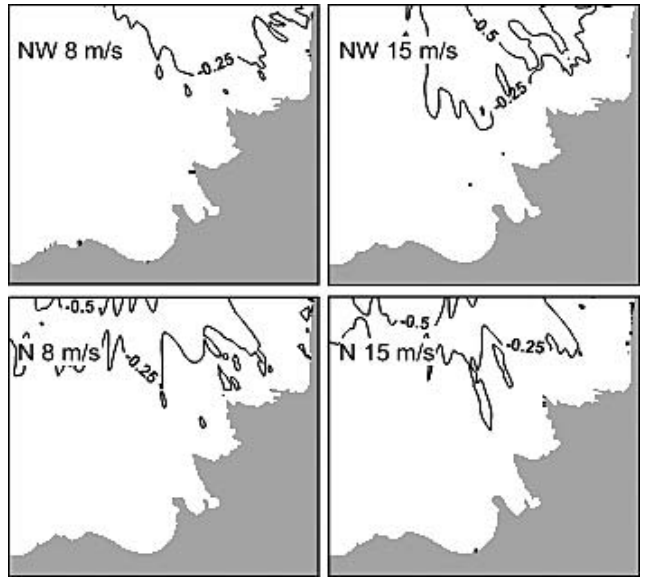

Figure 6. The impact of wind farm 2. The contour unit is [%] and a negative value shows a decrease of significant wave height.

The shortcoming of this simulation study and of the ones mentioned in the latter paragraph lies in the assumption that wind farms do not alter wind speed. In general this is not true (Christiansen and Hasager, 2006). However, establishing the exact loss of wind speed in a wind farm in the present situation needs a detailed study, which is not in the scope of this paper. In our opinion the changes in the wave fields due to a 10% loss of wind behind the farm established by Christiansen and Hasager (2006) will be superposed by the nearshore wave transformation processes.

Also, we have to emphasize here that although the presence of wind farms has a negligible effect on the wave field, the presence of vertical piles in water motion induces von Karman vortex shedding behind the structure. This vortex shedding provides some disturbances in the water column, particularly at the sea bottom, and may induce substantial effects on the local benthic communities at sea bottom.

CONCLUSIONS

A modeling study was undertaken in order to quantify the possible effects of offshore wind farms on the reduction of wave height near coast. The model was successfully verified against

the measurements at the locations where wind turbines would be installed. The impact of wind farms was assessed by calculating the ratio of difference of significant wave height with and without wind turbines to significant wave height without turbines. It is concluded that the reduction of significant wave height near the coast below 10-m isobaths does not exceed 1%. From a practical point of view, the developers of wind farms and the habitants/tourists on the coast should not be concerned.

ACKNOWLEDGMENTS

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